

PolarFire and Xilinx accelerator trade-off

Release 0.1 draft

LMO

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Status: DRAFT

PolarFire measurements are not available yet. Cells marked TBD are filled when the current inference campaign finishes.

Distribution:

- LMO (internal),
- European Space Agency (ESA)

Change Log:

- 0.1: First draft. Xilinx baseline taken from the CCN1 Design Justification File. PolarFire columns left open.

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2 Introduction

The DIOSSA RPO payload estimates the 6-degree-of-freedom pose of a target spacecraft from monocular images. The vision chain is two convolutional networks in series: FCOS for object detection, then MobilePose for keypoint regression. In the previous phase those networks were trained in floating point, refined with quantization-aware training, and compiled onto the Xilinx Deep Learning Processing Unit (DPU) in an UltraScale+ MPSoC through Vitis-AI 3.5.

This note compares that Xilinx baseline with the same networks compiled for the Microchip PolarFire SoC and its VectorBlox accelerator. The networks, the weights, and the task metrics stay the same. The item under test is the accelerator and its compile path.

2.1 Objective

Show that the CCN1 networks can move from the Xilinx DPU to PolarFire without a material loss of task accuracy, and define the measurements still required before that conclusion is closed.

2.2 Scope

The comparison uses the two networks that were embedded in CCN1 and that are being recompiled for PolarFire:

- Object detection: FCOS, input 384×288 .
- Keypoint regression: MobilePose, input 224×224 , 20 keypoints.

A second FCOS resolution, 640×480 , was trained in CCN1 and is reported here as context. It is not part of the PolarFire campaign.

2.3 Applicable Documents

AD-n	Document Title	Issue
AD-1	System Design Justification File (CCN1), <code>report/source.rst</code> in this repository	1.1

2.4 Definitions

Acronym	Meaning
DPU	Xilinx Deep Learning Processing Unit
EDGE	The network after compilation, running on the accelerator
FLOAT	The float32 network, evaluated on a GPU
FCOS	Fully Convolutional One-Stage object detector
IoU	Intersection over union of the predicted and true box
mAP	Mean average precision
QAT	Quantization-aware training; int8-constrained weights, evaluated in the training framework before compilation

3 What is being compared

Three representations of each network are kept separate, because they answer different questions.

Table 1: Representations of each network

Stage	What it is	What a difference means
FLOAT	Best float32 checkpoint from CCN1, run on a GPU	Training quality. Independent of the FPGA.
QAT	Same checkpoint after quantization-aware training, still evaluated in PyTorch	Accuracy given up to make the network representable in int8.
EDGE	The QAT network compiled and executed on the accelerator (Xilinx DPU, or PolarFire VectorBlox)	Accuracy and latency of the flight-like implementation.

FLOAT and QAT are properties of the checkpoint. They are identical for both targets, and they are taken from AD-1. EDGE is the only stage that can differ between Xilinx and PolarFire.

The PolarFire builds use the CCN1 QAT checkpoints directly:

- FCOS `ccn1--qu岸e-delf-epoch19`, input $1 \times 3 \times 288 \times 384$.

- MobilePose ccn1--bijou-rasp-epoch17, input $1 \times 3 \times 224 \times 224$, 20 keypoints.

4 Xilinx baseline

All figures in this section are copied from AD-1. They were measured on the CCN1 validation set unless noted otherwise. Inference rates quoted from AD-1 were measured on a laptop RTX 3070, not on the DPU. AD-1 does not publish a DPU frame rate.

4.1 Object detection (FCOS)

FCOS was selected over Faster R-CNN in CCN1. On a 480×640 input the float FCOS reached IoU 0.874 and mAP at IoU 0.75 of 0.956, against 0.853 and 0.901 for Faster R-CNN, at 45.78 FPS against 37.66 FPS on the RTX 3070, with 32.1 M parameters against 43 M. Faster R-CNN also cannot be trained with the Vitis-AI quantization-aware procedure, so it was dropped.

The embedded resolution is 384×288 . The larger 640×480 float model is more accurate (IoU 0.958 against 0.925) and about 30% slower on the GPU (46.13 FPS against 65.12 FPS). Quantization-aware training on 640×480 could not be run for long enough: the batch size was limited to 2 and the run lasted 47.7 minutes, finishing at IoU 0.819. The 384×288 QAT run used batch size 7 for 11.4 hours and finished at IoU 0.858. That is the network compiled for the edge.

Table 2: FCOS, 384×288 . Validation metrics from AD-1.

	FLOAT licit-weal	QAT quare-delf
Started from	COCO	licit-weal
Input [W × H]	384×288	384×288
IoU	0.925	0.858
mAP	0.858	0.693
mAP at IoU > 0.50	0.989	0.978
mAP at IoU > 0.75	0.946	0.850
mAP, large (area > 9216 px)	—	0.746
mAP, medium	—	0.691
mAP, small (area < 1024 px)	—	0.623
RTX 3070 throughput	65.12 FPS	not reported

The accepted quantization loss on this network is 0.067 IoU (0.925 down to 0.858) and 0.096 mAP at IoU 0.75 (0.946 down to 0.850). Detection at the loose threshold barely moves: mAP at IoU 0.50 goes from 0.989 to 0.978.

Table 3: FCOS, 640×480 . Reported for context. Not in the PolarFire campaign.

	FLOAT saved-aril	QAT livid-wire
Started from	COCO	saved-aril
Input [W × H]	640×480	640×480
IoU	0.958	0.819
mAP	0.939	0.658
mAP at IoU > 0.50	0.990	0.944
mAP at IoU > 0.75	0.988	0.826
RTX 3070 throughput	46.13 FPS	not reported

The 640×480 QAT model loses 0.139 IoU. After that loss it is less accurate than the 384×288 QAT model (0.819 against 0.858). The resolution that was actually embedded for the shorter approach is the smaller one.

4.2 Keypoint regression (MobilePose)

The float model selected in CCN1 is blest-har1, initialised from the DIOSSA-1 MobilePose checkpoint and trained on 20 keypoints. AD-1 states that raising the input from 224×224 to 448×448 did not improve accuracy and did increase training time, and that predicting all 41 keypoints was clearly worse than the 20-keypoint subset. The float experiment table itself is a figure in AD-1 (tables/big-kr-experiments-table.png) and is not in this repository, so the float row below contains only the two values written in the prose.

blest-har1 is reported at 3.799 L2 pixel error. The prose does not repeat whether that figure is on the 2590×1942 camera image or on the network input. The QAT section does give the float reference for L1 on the camera image: 4.783 px, against 5.303 px for the best QAT model.

The QAT model is `bijou-rasp`, started from `blest-har1`. It is the checkpoint compiled for the edge (`ccn1--bijou-rasp-epoch17`).

Table 4: MobilePose, 20 keypoints. Metrics from AD-1.

	FLOAT <code>blest-har1</code>	QAT <code>bijou-rasp</code>
Started from	DIOSSA-1 MobilePose	<code>blest-har1</code>
Input	224×224 (selected over 448×448)	224×224
Keypoints	20	20
L2 pixel error (reference frame not stated in the prose)	3.799	—
L1 on the camera image [2590×1942], px	4.783	5.303
L1 on the network input, px	not stated	1.564
L2 on the camera image [2590×1942], px	not stated	4.216
L2 on the network input, px	not stated	1.238

The accepted quantization loss is 0.520 px L1 on the camera image (4.783 to 5.303). Two other QAT learning rates were worse (`smoky-quey` 6.848 px, `swish-mows` 5.797 px), so `bijou-rasp` is the CCN1 selection.

4.3 Compiled models on the Xilinx DPU

AD-1 confirms that both networks compile with Vitis-AI and run on the DPU. The embedded pairs used for the trajectory evaluation are:

- 20 m to 100 m: `edge-bijou-rasp` with `edge-quare-delf`.
- 20 m to 300 m: `edge-bijou-rasp` with `edge-livid-wire`.

AD-1 does not publish a separate IoU or keypoint error for those compiled binaries. The task metrics above are the PyTorch QAT evaluation. The edge result in AD-1 is the end-to-end pose, given as image tables rather than text, plus a qualitative check that the DPU boxes and keypoints lie close to the ground truth on a sample frame.

What the prose does state:

- On the float pair, relative position error stays under 2% at 1-sigma on trajectories 1, 2, 3, 4 and 5, with good behaviour also on trajectories 4 and 7.
- On the QAT pair running on the DPU, relative position error stays under 2% at 1-sigma on trajectories 2 and 3.
- Trajectory 1 shows a rotation error near 180 degrees. AD-1 attributes that to symmetry of the chosen keypoints, not to quantization. The float models already place the position correctly and fail the rotation the same way.

Those pose tables are the right place to judge the DPU, and they are not transcribed in AD-1. Reading the figures back into this note is the remaining Xilinx-side action.

5 PolarFire measurements

The same QAT checkpoints are compiled to VNNX for the VectorBlox core. Post-processing that the core does not absorb runs as ONNX on the PolarFire SoC RISC-V cores. No accuracy or latency figure from that path is available yet.

Table 5: Edge comparison. PolarFire column to be filled from the current runs.

	Xilinx DPU	PolarFire VectorBlox
FCOS checkpoint	<code>quare-delf</code>	<code>quare-delf</code> , epoch 19
FCOS input	384×288	384×288
FCOS IoU on the shared sample	not tabulated in AD-1	TBD
FCOS mAP at IoU > 0.75	not tabulated in AD-1	TBD
FCOS accelerator latency	not reported in AD-1	TBD
FCOS end-to-end latency, including the CPU post-processing	not reported in AD-1	TBD
MobilePose checkpoint	<code>bijou-rasp</code>	<code>bijou-rasp</code> , epoch 17
MobilePose input	224×224 , 20 keypoints	224×224 , 20 keypoints
MobilePose L1 on the camera image, px	not tabulated in AD-1	TBD
MobilePose L2 on the network input, px	not tabulated in AD-1	TBD
MobilePose accelerator latency	not reported in AD-1	TBD
MobilePose end-to-end latency	not reported in AD-1	TBD

The sample used for the PolarFire runs is the CCN1 Inmarsat-5 test sample already staged for these two input sizes. The Xilinx edge numbers, once read out of the AD-1 pose tables or re-measured, have to be computed on that same sample. A comparison against the full CCN1 validation set would mix two test distributions.

6 How the result will be judged

The switch is supported if both of the following hold on the shared sample.

6.1 Task accuracy

The PolarFire edge network stays at or above the CCN1 QAT validation score, within the quantization loss that ESA already accepted when the DPU path was approved:

- FCOS at 384×288 : IoU at least 0.858, or a documented shortfall smaller than the 0.067 already lost between `licit-weal` and `quare-delf`.
- MobilePose: L1 on the 2590×1942 image at most 5.303 px, or a documented excess smaller than the 0.520 px already lost between `blest-har1` and `bijou-rasp`.

A gap inside that band is the same kind of loss as the quantization step, not a new loss introduced by PolarFire. A gap larger than that band has to be explained from the compile log (unsupported operators moved to the RISC-V head, scale and zero-point handling, or a difference in the test sample) before any claim is made.

These thresholds are the CCN1 validation-set scores. The PolarFire campaign runs on a smaller sample, so the numerical test is the difference between Xilinx edge and PolarFire edge on that sample, not a direct comparison of the sample score with 0.858 or 5.303.

6.2 Latency

AD-1 gives no DPU latency, so a numerical latency requirement cannot be taken from the previous file. The RPO chain is detection, then keypoint regression, then the pose solver. The budget that matters is the sum of the two accelerator times and the two CPU post-processing times, at the image rate the payload is required to close. That rate is not stated in AD-1 and has to be taken from the payload specification when the PolarFire times are in.

7 Open items

1. Transcribe the AD-1 edge pose tables (the third row of each table, which is the DPU evaluation of `edge-bijou-rasp` with `edge-quare-delf` and with `edge-livid-wire`) so the Xilinx edge column is numeric.
2. Confirm the reference frame of the `blest-har1` L2 figure of 3.799 px from the float experiment figure in AD-1.
3. Fill the PolarFire column: IoU, mAP, L1, L2, accelerator time, and end-to-end time, on the staged CCN1 sample, for `quare-delf` and `bijou-rasp`.
4. Re-measure the Xilinx DPU on that same sample, or show that the AD-1 edge tables already cover it.